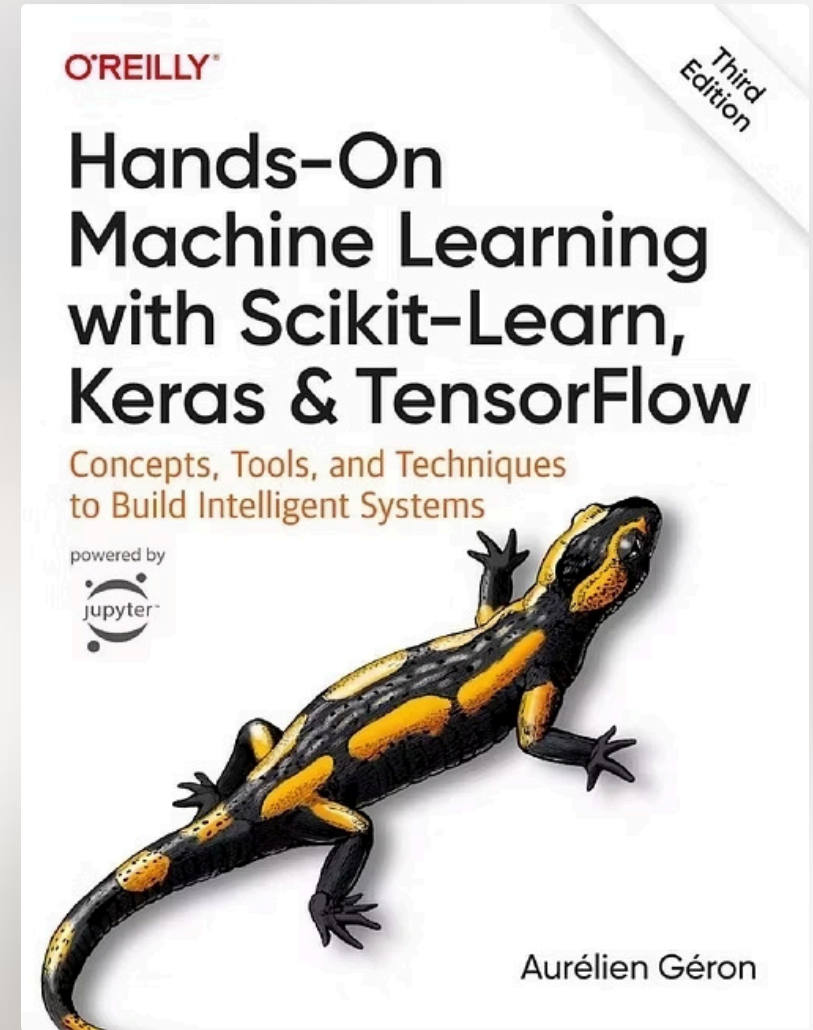


Machine Learning

Lessons based on the Textbook "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, 2023.

RP by Rodrigo Martins Pagliares

Last update: 07/08/2025



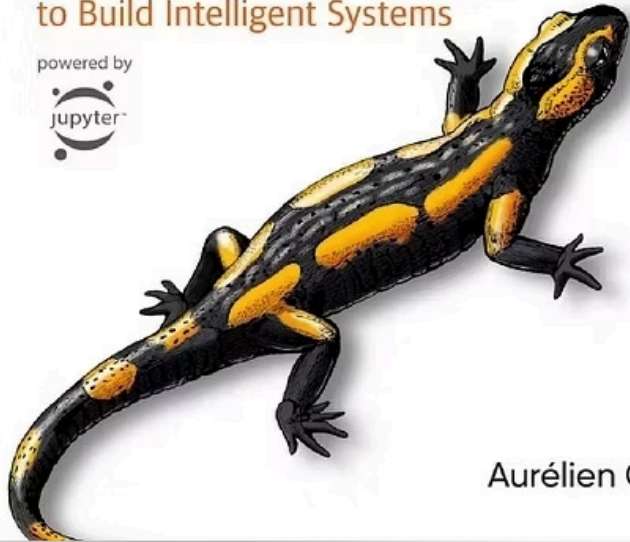
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Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow

Concepts, Tools, and Techniques
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Aurélien Géron

Machine Learning

Chapter 1 – The Machine Learning Landscape

Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
- 4. Types of Machine Learning Systems**
5. Main Challenges of Machine Learning
6. Testing and Validating





Agenda for Chapter 1

1. What Is Machine Learning?

2. Why Use Machine Learning?

3. Examples of Applications

4. Types of Machine Learning Systems

- Classification criteria #1: Supervision during training
 - Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement Learning
- Classification criteria #2: Approach to Learning Method
 - Batch and **Online (Incremental) Learning**
- Classification criteria #3: Learning Method
 - Instance-Based and Model-Based Learning

5. Main Challenges of Machine Learning

- Insufficient Quantity of Training Data
- Non-representative Training Data
- Poor-Quality Data
- Irrelevant Features
- Overfitting the Training Data
- Underfitting the Training Data
- Stepping Back

6. Testing and Validating

- Hyperparameter Tuning and Model Selection
- Data Mismatch

Classification criteria #2:

Approach to Learning

Method

Batch and Online (Incremental) Learning

Online Learning: Overview

- **System learns incrementally**
 - Processing data **sequentially**
 - individual instances or mini-batches.
- **Each learning step is fast and cheap,**
 - Allowing the system to **learn as new data arrives**.



Adapt to changing data on the fly

- Key Feature of Online (incremental) Learning.

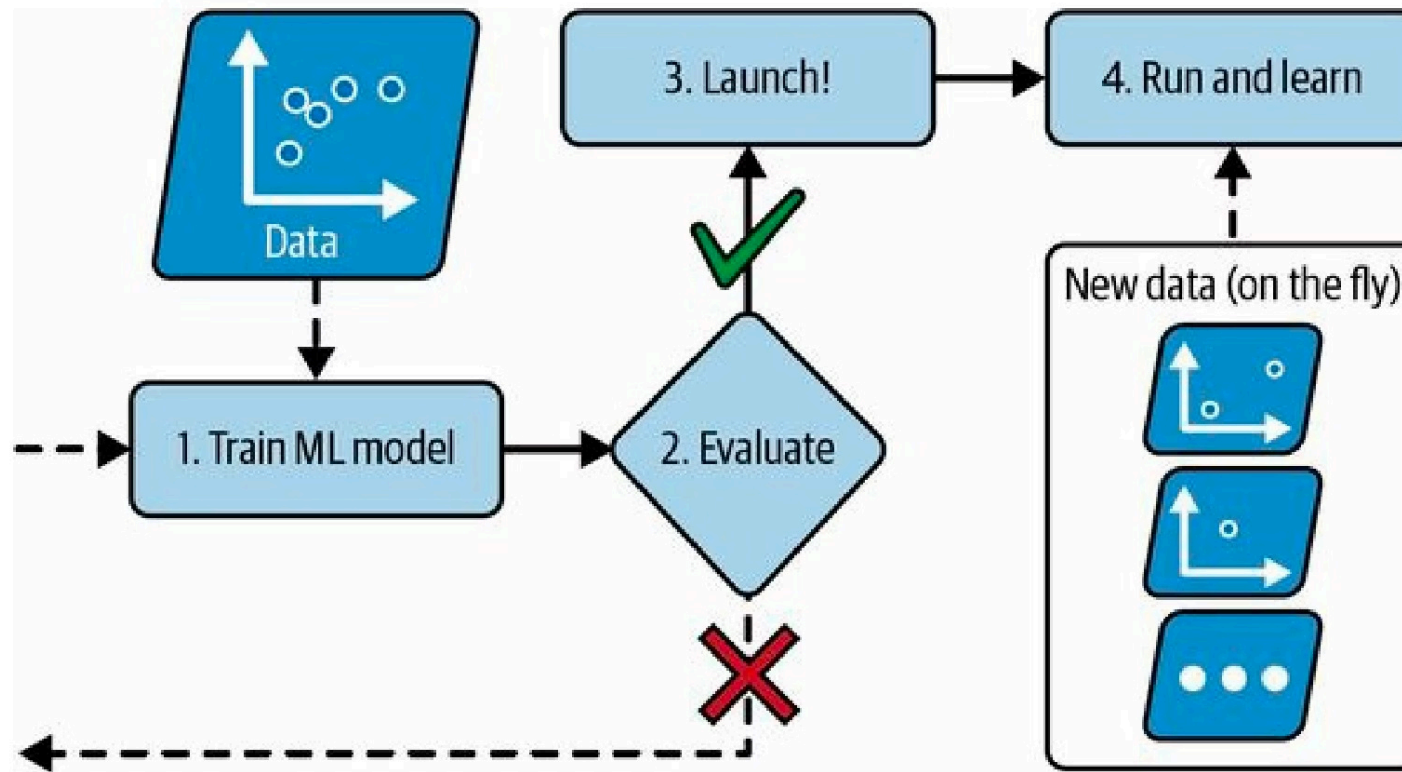


Figure 1-14. In online learning, a model is trained and launched into production, and then it keeps learning as new data comes in

Advantages of Online (Incremental) Learning

Rapid Adaptation:

- Useful for systems that need to adapt quickly to new patterns (e.g., **stock market predictions**).

Limited Computing Resources:

- Ideal for systems with **constrained resources**, such as **mobile devices**.

Huge Datasets:

- Enables **training on datasets** that do not fit in a single machine's memory (**out-of-core learning**).





Out-of-Core Learning



Load Data Batch

System loads a portion of the massive dataset that fits in available memory



Train Model

The learning algorithm trains on the loaded batch of data



Move to Next Batch

Process repeats until the entire dataset has been processed

- Benefit:

- Can handle massive datasets that exceed available memory.

Online Learning for huge datasets

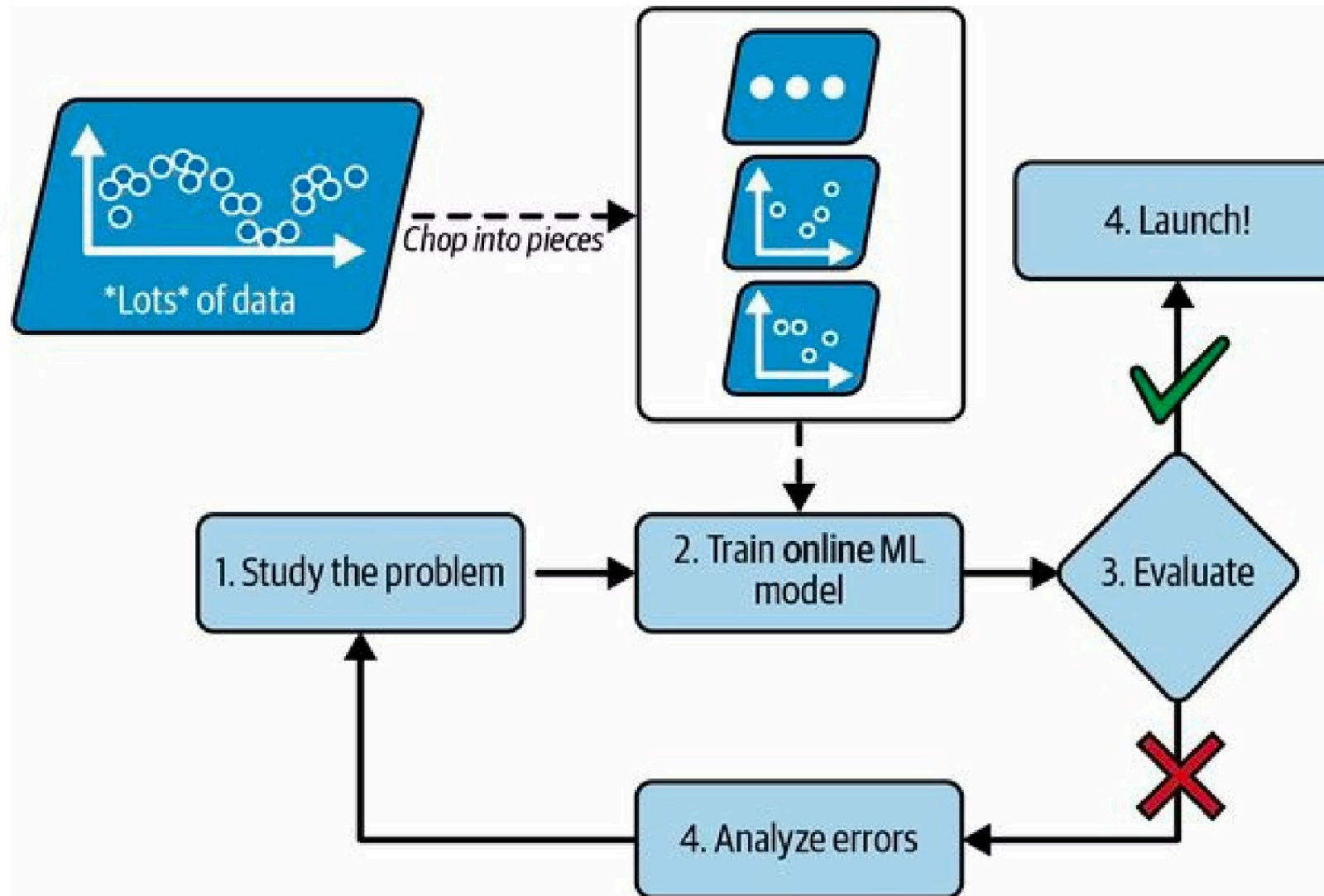
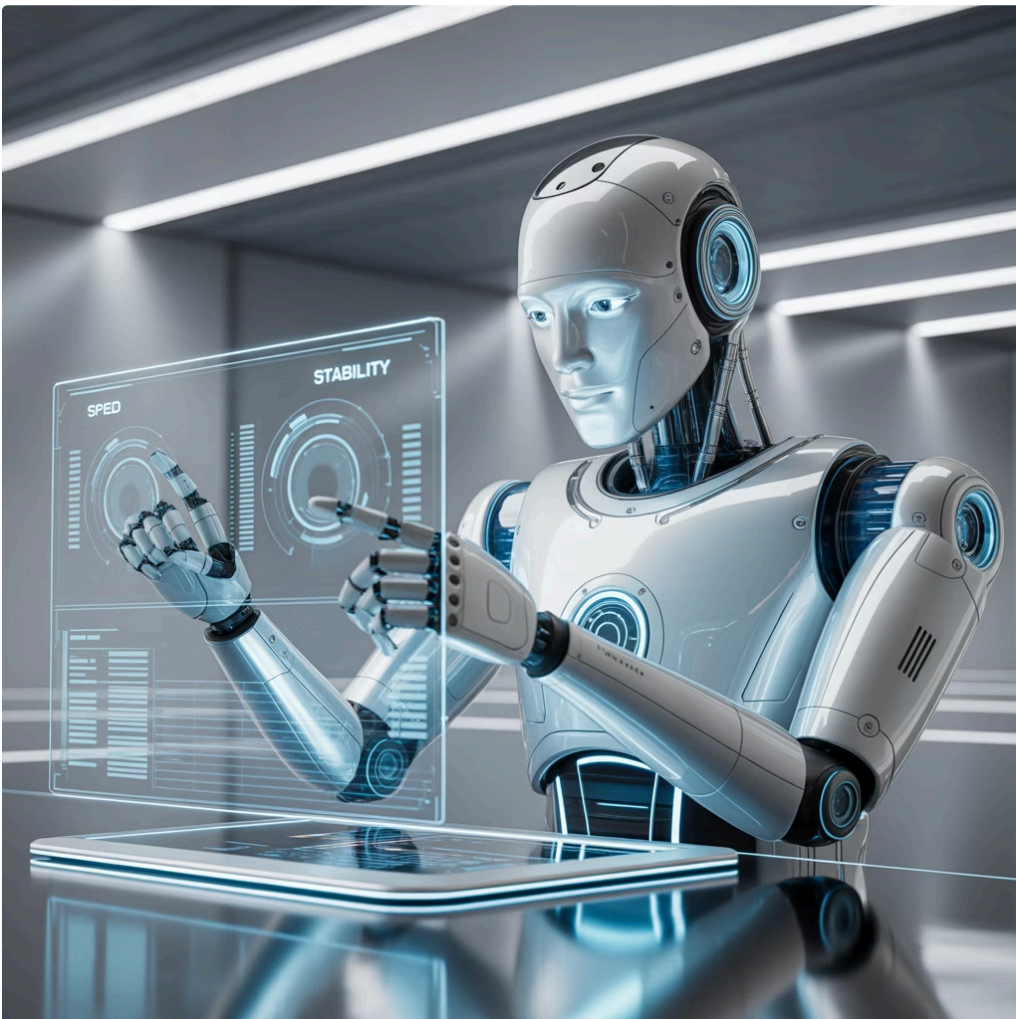


Figure 1-15. Using online learning to handle huge datasets

Learning Rate: Trade-Offs

- The Learning rate of online learning systems indicates how fast they should adapt to changing data.
 - Learning rate is an important parameter for online (incremental) systems.



1 High Learning Rate:

- Rapid adaptation to new data, but may forget old data quickly.
- Example: A spam filter that flags only the latest types of spam.

2 Low Learning Rate:

- Slower adaptation but more resistant to noise and outliers in the data.

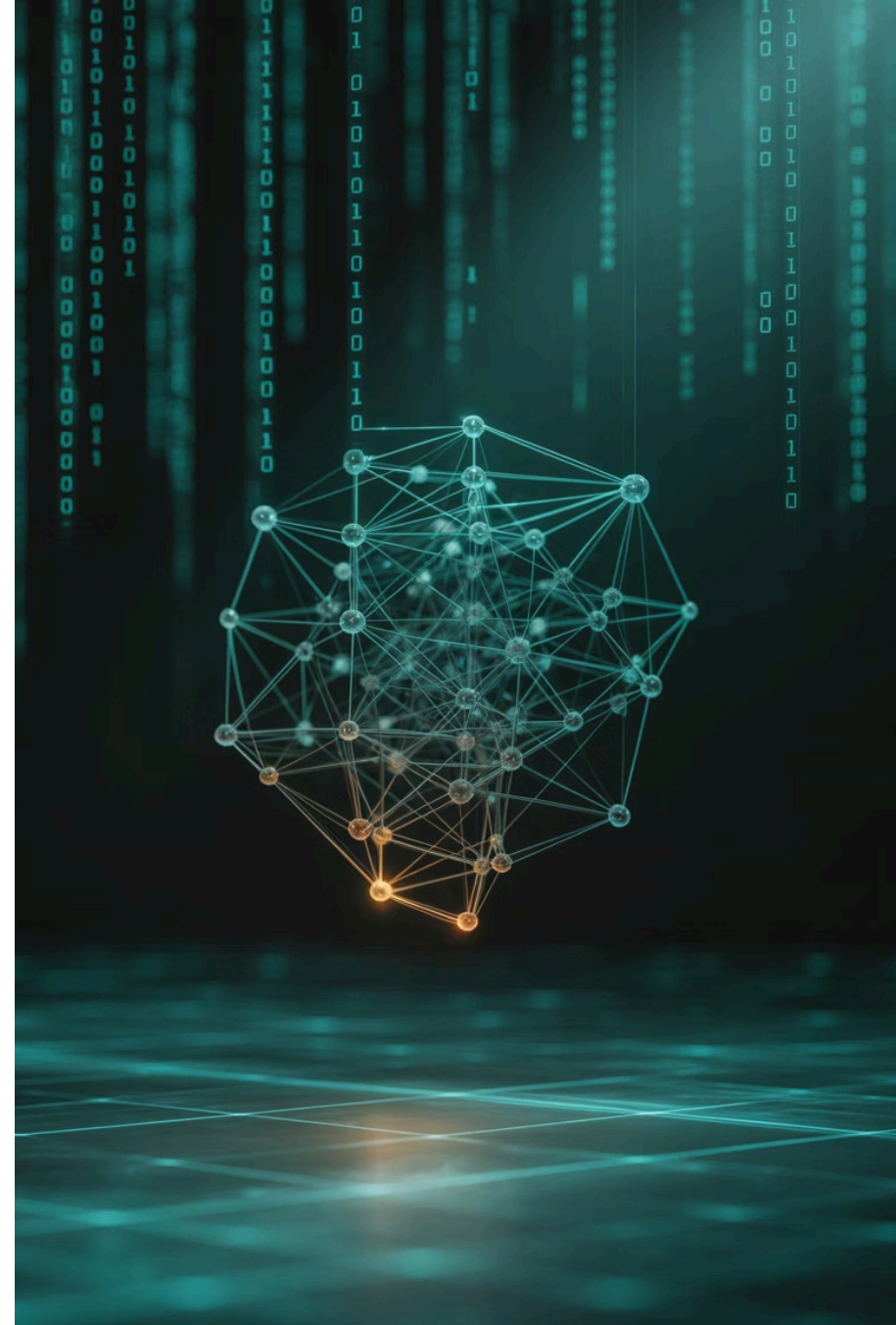


ChatGPT moment! (not in the textbook)

Can you better explain the concept of "forgetting" associated with learning rate for online (incremental) systems?

"Forgetting" in Online Learning Systems

- Forgetting refers to a **model losing significance on previously learned data**
- The **learning rate** controls how much the model adapts to new data
- **Forgetting** occurs as the **model prioritizes new data over old data**





High Learning Rate and "Forgetting"

- High learning rate leads to **significant weight updates**
- Model **prioritizes new data over older data**
- Previous **patterns** may be **overwritten** or **forgotten**
- Fast adaptation, but risk of forgetting older information

Low Learning Rate and Forgetting

- Low learning rate results in **smaller** updates to the model
- Model **retains** more **old data** while **gradually incorporating new** information
- Maintains **balance between new and old data**
- **Reduces the chance of forgetting** previously learned patterns





The Importance of "Forgetting" in Applications

- Forgetting impacts tasks like **spam filtering**
- **Model must learn new patterns without losing knowledge of old ones**
- **Balance is essential** to avoid misclassification of new data

Challenges & Considerations



Bad Data

Online learning systems are **vulnerable to bad data** (e.g., from a malfunctioning sensor or spam).



Monitoring

Systems should be **monitored to detect performance drops or abnormal data**, and learning may need to be paused or reverted.



Use of Anomaly Detection

Anomaly detection algorithms can be used to **identify and mitigate the effects of bad data**.



Summary of Online (Incremental) Learning

1 Adaptability:

Ideal for systems needing fast and continuous adaptation.

2 Efficiency:

Suitable for systems with limited resources or very large datasets.

3 Risk Management:

Close monitoring of data quality and system performance is crucial to prevent issues with noisy or incorrect data.



Next: Instance-Based Learning



Q&A



Open Floor

Open floor for discussion or clarifications



Discussion

Share your thoughts and questions about AI concepts